State inputs to the ECEC simulator

Anna Rachwalski

30 September 2026

This memo summarizes how I build the state-level inputs to the early childhood education and care (ECEC) simulator, and updates the 23 September memo. The simulator runs each state on its own, with three kinds of state input:

- Annual hours of care in six care types, for the base year 2019.
- The base-year childcare wage, where the wage curve starts.
- An employment-rate path by sex and age group.

The memo shows results for Kentucky (KY), New Jersey (NJ), New York State (NYS), New York City (NYC), and Virginia (VA). The simulator selects households by state, so NYC isn't yet a standalone run.

Changes since the 23 September memo

- The build now runs for all 50 states and DC.
- The targets include children in care under 10 hours a week.
- Hours use the simulator's 10- and 40-hour weeks, not NSECE means.
- Head Start slots become enrolled children, minus the pre-K overlap.
- The base-year wage and the employment-rate path are state-level.

The simulator

A state run uses the state's ACS households, re-anchors demand to the state's targets, and reads the state's base-year wage and employment-rate path. Re-anchored hours replace national hours everywhere, including in base-year labor. All other inputs stay national.

Two things define each care type: the setting of the care, and whether the household pays for it. Paid means the household reports a positive net weekly cost for that arrangement.

The six care types, by setting and payment

Setting	Household pays	Household doesn't pay
Center	Center-Based	Unpaid Center-Based
Home: individual care outside the child's home, no prior relationship	Paid Home-Based	Unpaid Home-Based
Other: individual care in the child's home, or with a prior relationship	Other Paid	Other Unpaid

Formal care is Center-Based, Unpaid Center-Based, and Paid Home-Based. Informal care is the other three types. A child is in regular care when the child has at least five hours a week of non-parental, non-school care, the NSECE definition. A child with more than one type of care gets one primary type, which is the first type in this order: high-priced Center-Based, low-priced Center-Based, Unpaid Center-Based, Paid Home-Based, Other Paid, Unpaid Home-Based, Other Unpaid. Both price bands go into the one Center-Based type. The child's hours are the hours in the primary type.

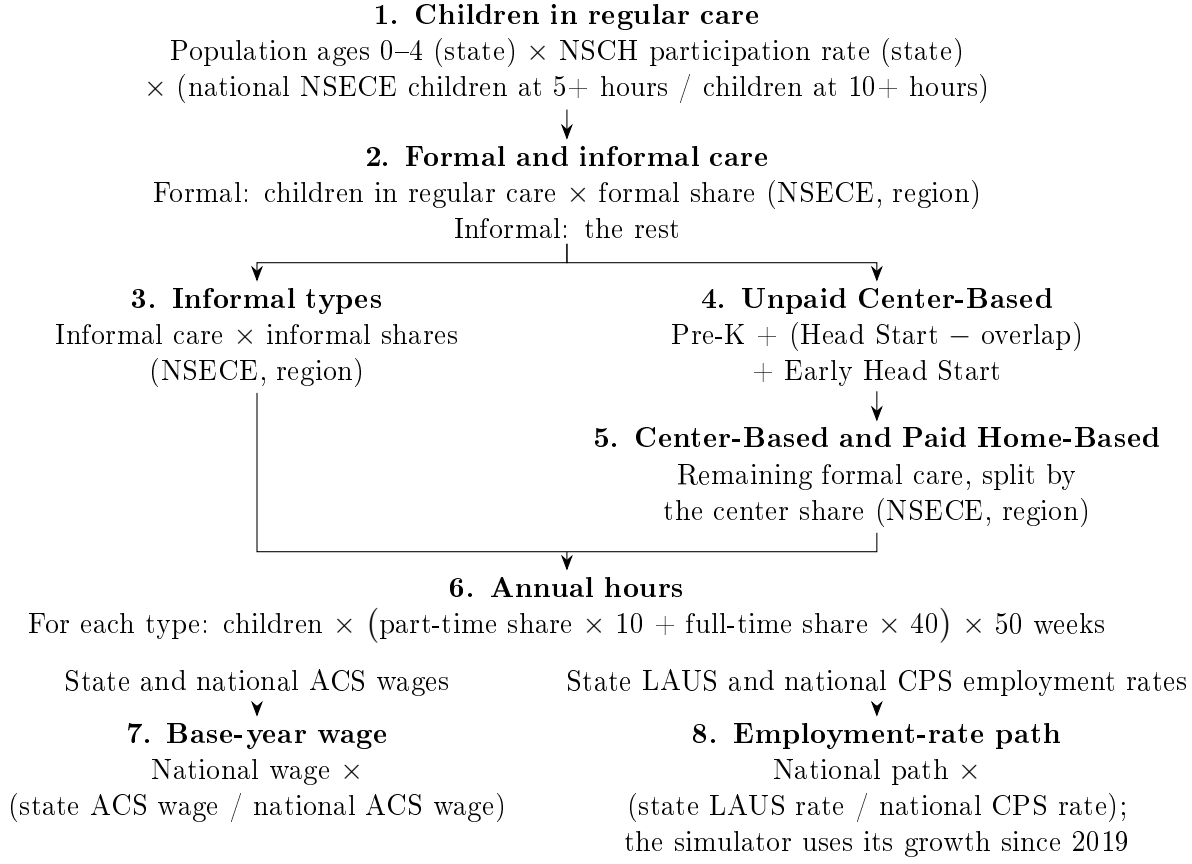
Data

The National Survey of Children's Health (NSCH) gives the share of children in non-parental care, by state (step 1). The National Survey of Early Care and Education (NSECE) gives the type of care and its hours (steps 1–6). State program counts give public pre-K, Head Start, and Early Head Start (step 4). The American Community Survey (ACS) gives childcare wages (step 7), and the BLS Local Area Unemployment Statistics give state employment rates (step 8).

Every NSECE constant in the demand build comes from the NSECE 2019 household file. The universe ratio (step 1) is national. The other constants are by Census region (steps 2–3 and 5–6). When a region has fewer than 30 unweighted NSECE records for a care type, I use the national value. This rule applies only to Unpaid Home-Based. States in the same region differ in their demand targets only through population, NSCH rate, and program counts.

The build

The figure shows the build's steps. The table gives the demand targets the build writes.



Annual hours by care type, 2019 (millions)					
Care type	KY	NJ	NYS	NYC	VA
Center-Based	73.1	194.4	428.5	202.3	151.7
Unpaid Center-Based	67.0	82.0	149.9	66.2	71.1
Paid Home-Based	18.6	29.5	64.9	30.7	38.7
Other Unpaid	76.2	171.1	358.8	166.7	123.7
Other Paid	54.9	77.8	163.1	75.8	89.2
Unpaid Home-Based	5.3	9.5	19.9	9.3	8.7

The NYS results include NYC. To get the rest of the state, subtract NYC rows from NYS rows.

The demand targets

Step 1. Children in regular care

The control total is the number of children 0–4 in regular care in the state. The population is from the 2019 Census population estimates.

$$\text{control total} = \text{population 0–4} \times \text{NSCH participation rate} \times \text{universe ratio} \quad (1)$$

- Rate: the weighted share of children 0–4 who get care for at least 10 hours a week from someone other than a parent or guardian (NSCH K6Q20, 2019). The rate is 59.4% in KY, 60.2% in NJ, 57.7% in NYS, and 52.0% in VA. NYC uses the NYS rate.
- Universe ratio: 1.098. In the NSECE, the weighted number of children in regular care (five hours or more) is 1.098 times the number whose arrangements add up to 10 hours or more, the NSCH threshold. I assume the ratio is the same in each state.

Step 2. Formal and informal care

$$\text{formal} = \text{control total} \times \text{regional formal share} \qquad \text{informal} = \text{control total} - \text{formal} \quad (2)$$

The regional formal share is the weighted share of children in regular care whose primary type is formal. The formal share is 56.5% in the Northeast, 52.9% in the Midwest, 53.2% in the South, and 48.3% in the West.

Step 3. Informal types

The build divides informal care among the three informal types by each type's regional share of weighted children with an informal primary type. Informal care accounts for 43–47% of the estimated children in regular care and 46% of the modeled annual hours. The table gives the informal shares, and the part-time shares of step 6.

Informal and part-time shares, NSECE, 2019								
	Northeast		Midwest		South		West	
Care type	Informal	<20 hrs	Informal	<20 hrs	Informal	<20 hrs	Informal	<20 hrs
Center	—	0.264	—	0.367	—	0.195	—	0.197
Unpaid Center	—	0.333	—	0.351	—	0.273	—	0.394
Paid Home	—	0.146	—	0.116	—	0.041	—	0.034
Other Unpaid	0.655	0.156	0.570	0.378	0.569	0.262	0.656	0.372
Other Paid	0.310	0.203	0.395	0.418	0.396	0.223	0.309	0.336
Unpaid Home*	0.035	0.115	0.035	0.115	0.035	0.115	0.035	0.115

*National value (5–14 unweighted NSECE child records per region); I rescale the other shares so informal adds to 1.

Step 4. Unpaid Center-Based

The build sets Unpaid Center-Based to the children in public pre-K, Head Start, and Early Head Start, minus the Head Start children pre-K already counts. The rest of formal care goes to step 5:

$$\text{Unpaid Center-Based} \;=\; \text{pre-K} + \big(\text{Head Start} - \text{overlap}\big) + \text{Early Head Start} \tag{3}$$

$$\text{other formal} \;=\; \text{formal} - \text{Unpaid Center-Based} \tag{4}$$

This assumes no program child also uses a paid center or is in other free center care. Public pre-K is the pre-K grade enrollment in the Common Core of Data (CCD) of the National Center for Education Statistics (NCES), on or near October 1, 2018. It counts grade, not age, so it includes some five-year-olds.

Estimated children in public programs, 2018–19					
	Public pre-K	Head Start	Est. overlap	Early Head Start	Total
KY	28,465	11,781	−798	2,673	42,121
NJ	41,305	11,430	−1,300	3,183	54,618
NYS	54,451	35,288	−72	10,171	99,838
NYC	26,717	12,149	0	5,241	44,107
VA	33,790	11,084	−2,385	2,230	44,720

4a. Head Start and Early Head Start

The state sources give funded slots, not children. Head Start slots come from the National Institute for Early Education Research (NIEER) 2019 yearbook, for ages 3 and 4 (NIEER counts five-year-olds as four). Early Head Start slots come from a NIEER report, for children under 3.

Slots and children differ: some children leave, and some Early Head Start slots are for pregnant women. I multiply each state's funded slots by a national number of children per funded slot: 0.957 for Head Start and 0.890 for Early Head Start.

$$0.957 \;=\; \frac{662{,}963 \text{ children enrolled at year end}}{692{,}546 \text{ funded slots for children}} \qquad 0.890 \;=\; \underbrace{\frac{153{,}650}{166{,}505}}_{0.923} \times \underbrace{\left(1 - \frac{5{,}988}{166{,}693}\right)}_{0.964} \tag{5}$$

All figures are national, from the Head Start Program Information Report (PIR), 2018–19. Year-end enrollment is cumulative enrollment minus the children who left and weren't replaced. I treat the year-end Head Start count and the October pre-K count as one year.

For Early Head Start, the 0.964 factor removes the slots for pregnant women: 5,988 of 166,693 national funded slots. The 0.923 factor then converts the child slots to children. The slot totals for children also include the American Indian and Alaska Native (AIAN) child slots. I split these between Head Start and Early Head Start by the ages of AIAN enrolled children.

NYC's Head Start and Early Head Start counts are the city's share of the NYS counts, from the funded slots at open sites in the Head Start site file of the Administration for Children and Families (ACF). The only vintage of that file is a 2026 export. I assume the share was the same in 2019.

4b. Overlap

The sum of the three programs counts some children twice. CCD pre-K excludes Head Start students "unless Head Start is part of an authorized public education program of an LEA" (local education agency; CCD glossary). A Head Start class that a school district runs can be in both CCD pre-K and the federal Head Start count.

I estimate this overlap from the ACF site file, CCD pre-K membership by school, the CCD school directory, and the NCES school geocodes. I flag funded slots as overlap when the site passes all three tests:

1. The site is at a school with pre-K: the address matches, or the two geocodes are within 150 meters.
2. The agency that runs the site has a school district's name, such as "public schools."
3. The agency and the school's district share a place name.

$$\text{overlap} \;=\; \text{Head Start children (4a)} \times \text{flagged share of Head Start slots} \tag{6}$$

The flagged share is 6.8% in KY, 11.4% in NJ, 0.2% in NYS, 0.0% in NYC, and 21.5% in VA. This is an estimate: the site file is a 2026 export, and the tests compare names and locations but don't confirm that a child is in both counts. Without the deduction, Unpaid Center-Based hours are at most 5.3% higher (VA), Center-Based and Paid Home-Based hours at most 2.2% lower, and total hours change by less than 0.1%.

Step 5. Center-Based and Paid Home-Based

Center-Based is other formal care times the regional center share, and Paid Home-Based is the rest. The center share is the weighted share of Center-Based and Paid Home-Based children who are Center-Based. It is 0.880 in the Northeast, 0.697 in the Midwest, 0.816 in the South, and 0.810 in the West.

Step 6. Annual hours

The build uses the simulator's 10- and 40-hour weeks, for 50 weeks a year:

$$\text{annual hours} \;=\; \text{children} \;\times\; \big(\text{part-time share} \times 10 + \text{full-time share} \times 40\big) \;\times\; 50 \tag{7}$$

The part-time share is the weighted share of children of a primary type with under 20 hours a week in that type. The file holds only the six totals, so the simulator can match a target with other child counts or 10/40 mixes.

Base-year wage and employment-rate path

Step 7. Base-year wage

$$\text{base-year wage} \;=\; \text{national wage} \times \frac{\text{state ACS wage}}{\text{national ACS wage}} \tag{8}$$

The national wage is \$12.06 an hour for workers without a bachelor's degree and \$19.70 for workers with one. I use the ACS for the ratio, from the 2018 and 2019 records of the 2015–2019 ACS five-year sample. An education group with fewer than 50 unweighted ACS records keeps the national wage. The 2019 base-year wages, without and with a degree, are \$10.28 and \$16.15 in KY, \$13.63 and \$23.89 in NJ, \$13.40 and \$23.17 in NY, and \$11.92 and \$19.08 in VA.

Step 8. Employment-rate path

The simulator uses only the path's growth since 2019, applied to each employment group's 2019 rate from the state's households. For each year:

$$\text{state path} \;=\; \text{national path} \times \text{state ratio} \qquad \text{state ratio} \;=\; \frac{\text{LAUS state employment rate}}{\text{CPS national employment rate}} \tag{9}$$

The state rates are from the Local Area Unemployment Statistics (LAUS) of the Bureau of Labor Statistics: annual means of the monthly, seasonally adjusted rates for ages 16 and older, 2006–2024. LAUS is controlled to the Current Population Survey (CPS). LAUS has no sex or age detail, so the simulator gives every sex and age group the same state adjustment. A change in a state's age mix can move its ratio, so the ratio isn't a separate state trend for each group. The ratio keeps its 2024 value, the last LAUS year, so for every group and year of a run:

$$\text{state growth since 2019} \;=\; \text{national growth since 2019} \times \frac{\text{state ratio, 2024}}{\text{state ratio, 2019}} \tag{10}$$

The state adjustment, the last term, is 0.992 in KY, 1.002 in NJ, 1.011 in NY, and 1.003 in VA (0.960 to 1.043 across all 50 states and DC).

Next steps. Make NYC a standalone simulator run. Find out whether labor per hour, elasticities, and wage and price growth can be state-level.